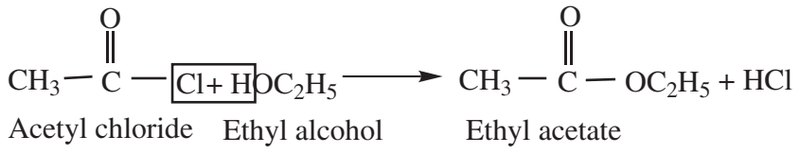
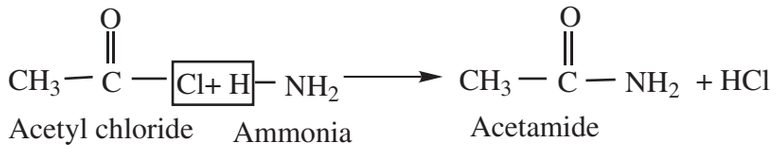
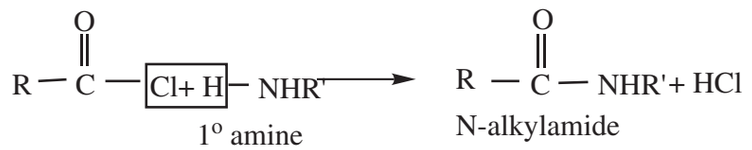
2) Reaction with Alcohols (*Alcoholysis*) gives esters.

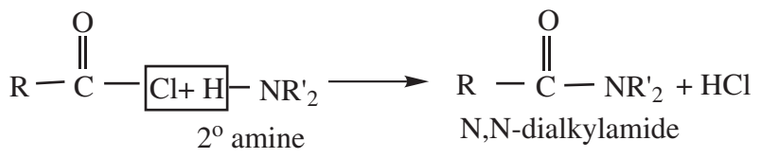


3) Reaction with Ammonia (*Ammonolysis*) gives acid amides.



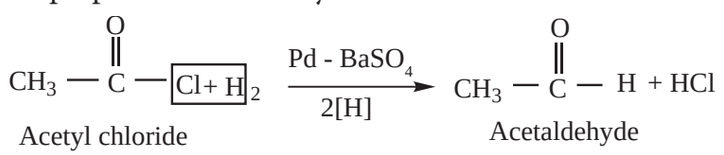
4) Reaction with 1° and 2° Amines gives N-alkyl amides.



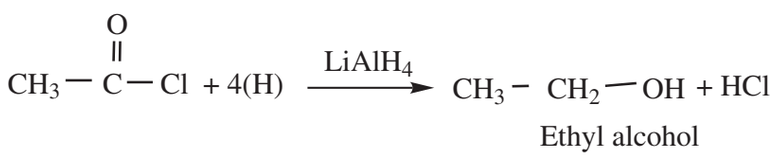


(5) Reduction.

(a) When reduced with hydrogen in the presence of 'poisoned' palladium catalyst, they form aldehydes. This reaction is called Rosenmund reduction. We have already learnt this reaction under the preparation of aldehydes



(b) When reduced with $LiAlH_4$ gives primary alcohols.



12.14.3 Acid anhydride

Methods of preparation

1. Heating carboxylic acid with P_2O_5

We have already learnt that when carboxylic acids are heated with P_2O_5 dehydration takes place to form acid anhydride.

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